

Some diagrammatic monoidalities

Only nodes:

$$A \otimes (B \otimes C) \qquad (B \otimes C) \otimes A$$

$$(A \otimes B) \otimes C$$

$$B \otimes (C \otimes A)$$

$$(B \otimes A) \otimes C \qquad B \otimes (A \otimes C)$$

With arrows and all trivialities:

$$\begin{array}{ccccc}
 & & A \otimes (B \otimes C) & \xrightarrow{\gamma} & (B \otimes C) \otimes A \\
 & \nearrow \alpha & & & \searrow \alpha \\
 (A \otimes B) \otimes C & & & & B \otimes (C \otimes A) \\
 & \searrow \gamma \otimes 1 & & & \nearrow 1 \otimes \gamma \\
 & & (B \otimes A) \otimes C & \xrightarrow{\alpha} & B \otimes (A \otimes C)
 \end{array}$$

A pentagon:

$$\begin{array}{ccc}
 & (A \otimes (B \otimes C)) \otimes D & \\
 \nearrow \alpha_{A,B,C} \otimes 1_D & & \searrow \alpha_{A,B \otimes C,D} \\
 ((A \otimes B) \otimes C) \otimes D & & A \otimes ((B \otimes C) \otimes D) \\
 \downarrow \alpha_{A \otimes B,C,D} & & \downarrow 1_A \otimes \alpha_{B,C,D} \\
 (A \otimes B) \otimes (C \otimes D) & \xrightarrow{\alpha_{A,B,C \otimes D}} & A \otimes (B \otimes (C \otimes D))
 \end{array}$$

You'll notice that I've used some phantom characters in the above diagram's label code. Try removing them to see what happens. Things like this are great little quick and dirty tricks to fix some common positional and layout problems.

A triangle:

$$\begin{array}{ccc}
 (A \otimes I) \otimes B & \xrightarrow{\alpha_{A,I,B}} & A \otimes (I \otimes B) \\
 \searrow \rho_A \otimes 1_B & & \swarrow 1_A \otimes \lambda_B \\
 & A \otimes B &
 \end{array}$$

Here is a version of the pentagon diagram that's been made too big to fit on the page:

$$\begin{array}{ccccc}
 & & (A \otimes (B \otimes C)) \otimes D & & \\
 & \nearrow \alpha_{A,B,C} \otimes 1_D & & \nwarrow \alpha_{A,B \otimes C,D} & \\
 ((A \otimes B) \otimes C) \otimes D & & & & A \otimes ((B \otimes C) \otimes D) \\
 \searrow \alpha_{A \otimes B,C,D} & & & & \swarrow 1_A \otimes \alpha_{B,C,D} \\
 (A \otimes B) \otimes (C \otimes D) & \xrightarrow{\alpha_{A,B,C \otimes D}} & A \otimes (B \otimes (C \otimes D)) & &
 \end{array}$$

Short of redesigning it so that it does fit, one may simply scale the image by encapsulating the entire diagram's code in a scale-box with a scale factor between 0 and 1. Here I've used 0.75:

$$\begin{array}{ccccc}
 & & (A \otimes (B \otimes C)) \otimes D & & \\
 & \nearrow \alpha_{A,B,C} \otimes 1_D & & \nwarrow \alpha_{A,B \otimes C,D} & \\
 ((A \otimes B) \otimes C) \otimes D & & & & A \otimes ((B \otimes C) \otimes D) \\
 \searrow \alpha_{A \otimes B,C,D} & & & & \swarrow 1_A \otimes \alpha_{B,C,D} \\
 (A \otimes B) \otimes (C \otimes D) & \xrightarrow{\alpha_{A,B,C \otimes D}} & A \otimes (B \otimes (C \otimes D)) & &
 \end{array}$$

In most situations this is not ideal (including this one in my opinion) and it is better to choose a different style of diagram. In fact it would be better to use xygraph or xy instead of xymatrix due to not being bound to the rigid matrix-entry-like positioning (i.e. in xy you can have nodes overlapping each other and, while you can force this in xymatrix, it is pretty hopeless).